Alonso Javier Bustos Barría[1]
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Date of birth: 19 February 2002 *Nationality:* Chilean

Education

2026 – Present	Ph.D. Student in Mathematics, George Mason University, Fairfax, Virginia, USA.
2020 – 2025	Mathematical Civil Engineering, Universidad de Concepción, Chile.
2020 – 2024	Bachelor of Sciences in Engineering Mathematics, Universidad de Concepción, Chile.

Scholarships and prizes

2026	Universidad de Concepción prize for the best graduate from the Mathematical Civil Engineering program.
2026	Best Graduate Award – Chilean Engineers Association (Colegio de Ingenieros de Chile). Awarded to top graduate of the program based on academic excellence and leadership.
2020	Universidad de Concepción scholarship for best national score in University Selection Test (PSU) in Mathematics.
2020	Mathematical engineering scholarship for the best 10 applicants to Mathematical Engineering at Universidad de Concepción.
2019	Awarded for achieving the highest national score in the University Selection Test (PSU) in Mathematics.

Academic Experience

2026	**Graduate Research Assistant** (Aug 2026 – May 2027), Department of Mathematical Sciences, George Mason University, under the supervision of Professor Harbir Antil.
2026	**Research Assistant** (Mar 2026 – Jun 2026), GIANuC[2], Universidad Católica de la Santísima Concepción and CMM: PIA Program FB210005, under the supervision of Professor Sergio Caucao.

[1]Latest update: October 22, 2026

2025 **Undergraduate Thesis Student of Mathematical Civil Engineering** (Aug 2025 – Dec 2025), Universidad de Concepción, under the supervision of Professor Sergio Caucao, supported by Fondecyt Project No. 1250937, "Numerical Analysis of Nonlinear Single and Coupled Problems with Biomedical and Geoscience Applications." The thesis is entitled "*A Banach Space Mixed Formulation for the Unsteady Brinkman Problem with Spatially Varying Porosity.*"

Research Stay (Feb 2025 – Mar 2025), George Mason University, Fairfax, Virginia, USA: A two-month research stay in February and March under the supervision of Professor Harbir Antil. The main goal was to investigate topics related to Risk measures and Rockafellians in the context of PDE-constrained optimization under uncertainty.

2024 **Research Assistant** (Jun 2024 – Dec 2024), CI^{2}MA and Universidad del Bío-Bío: Anillo of Computational Mathematics for Desalination Processes, Código ACT210087, under the supervision of Professor Gabriel N. Gatica.

Research Internship (Jun 2024 – Jul 2024), Universidad Católica de la Santísima Concepción: Fondecyt Project No. 11220393, "*Finite Element Methods for Brinkman–Forchheimer and Related Problems*", under the supervision of Professor Sergio Caucao.

Publications

Published

1. A. J. Bustos and S. Caucao, *A posteriori error analysis of two mixed formulations for a coupled Brinkman–Forchheimer and convection-diffusion-reaction system.* Journal of Computational and Applied Mathematics, accepted for publication.

2. A. J. Bustos, G. N. Gatica, R. Ruiz-Baier, and B. N. Venegas. *A perturbed threefold saddle-point formulation yielding new mixed finite element methods for poroelasticity with reduced symmetry.* ESAIM: Mathematical Modelling and Numerical Analysis, accepted for publication.

3. A. J. Bustos, S. Caucao, G. N. Gatica, and B. N. Venegas. *New fully mixed finite element methods for the coupled convective Brinkman–Forchheimer and nonlinear transport equations.* J. Sci. Comput. 104 (2025), Paper No. 64, 45 pp. (Available here).

4. A. J. Bustos, S. Caucao, and G. N. Gatica. *Mixed-primal and fully-mixed formulations for the convection-diffusion-reaction system based upon Brinkman–Forchheimer equations.* Comput. Math. Appl., vol. 207 (2026), pp. 222–249. (Available here).

Submitted

1. A. J. Bustos, J. J. Lee, and R. Ruiz-Baier. *Fully mixed finite element methods for the coupling of viscoelastic and reaction-diffusion models.* Preprint, 2026.

2. A. J. Bustos and S. Caucao. *A Banach space mixed formulation for the unsteady Brinkman problem with spatially varying porosity.* Preprint, 2025. (Preprint available here).

3. H. Antil, A. J. Bustos, S. P. Carney, and B. N. Venegas. *Rockafellian Relaxation for PDE-Constrained Optimization under Uncertainty in the context of Risk Measures.* Preprint, 2026. (Preprint available here).

In preparation

1. A. J. Bustos and S. Caucao. *A posteriori error analysis of a fully discrete mixed scheme for the unsteady Brinkman problem with spatially varying porosity.*

Teaching Experience

2026 **Part-Time Lecturer**, Universidad del Bío-Bío: Linear Algebra, and Algebra for engineering.

Teaching Assistant, Universidad Católica de la Santísima Concepción: Calculus I and Mathematical Foundations I.

2025 **Teaching Assistant**, Universidad de Concepción: Functional analysis I, and Mixed Finite Element Methods.

2024 **Teaching Assistant**, Universidad de Concepción: Functional Analysis I and Mathematical Thinking for Mathematical Engineering students. Ordinary Differential equations for engineering students.

Course Tutoring of Real Analysis I for mathematical engineering students.

2023 **Teaching Assistant**, Universidad de Concepción: Ordinary Differential Equations, Numerical Analysis I and Complemento de Cálculo (Complex Analysis and Partial Differential Equations) for engineering students. Algebra III: Foundations and Linear Algebra II for Mathematical Engineering students. Discrete Mathematics for Mathematics Pedagogy students.

2022 **Teaching Assistant**, Universidad de Concepción: Calculus I, Algebra II and Numerical Analysis I for Engineering students.

Course Tutoring of Calculus IV for Mathematical Engineering students.

2021 **Teaching Assistant**, Universidad de Concepción: Algebra I, Algebra II and Calculus I for Engineering students.

Talks and posters

2026 Finite Element Circus, Fall 2026, at University of Connecticut. *A posteriori error analysis for the unsteady Brinkman problem with varying porosity.*

2026 CIMPA School on Numerical Analysis and PDEs: Classical methods and recent developments, Rosario and Buenos Aires, Argentina. *On a stress-velocity-vorticity formulation for the unsteady Brinkman equations with variable porosity: A priori and a posteriori analyses.*

2025 Talca Numérica I, at the Universidad Católica del Maule. *Banach Space Mixed Formulations for the Convection-Diffusion-Reaction System Based upon Brinkman–Forchheimer Equations.*

2025 Seminar on Numerical Analysis and Mathematical Modeling at the Centro de Investigación en Ingeniería Matemática (CI^{2}MA). *Rockafellian Relaxation for PDE-Constrained Optimization under Uncertainty in the Context of Risk Measures.*

2025 Meeting of the Chilean Mathematical Society (SOMACHI), Vicuña, Chile. *A Banach Space Mixed Formulation for the Unsteady Brinkman Problem with Spatially Varying Porosity.*

Conferences and Workshops attended

2026 Finite Element Circus, Fall 2026, at University of Connecticut.

2026 CIMPA School on Numerical Analysis and PDEs: Classical methods and recent developments, Rosario and Buenos Aires, Argentina.

2025 Meeting of the Chilean Mathematical Society (SOMACHI).

2025 Talca Numérica I (at the Universidad Católica del Maule).

2024 Jornada Matemática de la Zona Sur (at the Universidad Católica de Temuco).

2024 Workshop on Numerical Analysis of Partial Differential Equations (WONAPDE 2024).

2022 Encuentro Nacional en Ingeniería Matemática (ENIM 2022).

Computer Skills

Programming Languages	Python
Mathematical Software/Libraries	MATLAB, FEniCS, FreeFem++, DUNE-FEM, NGSolve

Other Activities

2025 Member of the Organizing Committee of the *Encuentro Nacional de Ingeniería Matemática 2025* (National Meeting of Mathematical Engineering).

2024 Co-organizer of *Seminario de Análisis Numérico y Modelación Matemática* (Seminar on Numerical Analysis and Mathematical Modeling) at CI^{2}MA, Concepción, Chile.

2024 Member of the Electoral Board of Mathematical Engineering at Universidad de Concepción.